



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
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SDC 1-Day 10
II-II CSE, CSE(AI&ML)
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SL No	Week	Part A	Detailed Subtopics	Part B	Detailed Subtopics
1	Week 4	Principal Component Analysis (PCA)	Curse of Dimensionality, PCA theory and intuition, What Are Principal Components, Standardization, Covariance Matrix, Eigenvalues & Eigenvectors, Conclusion, Limitations	Practical Implementation	Day10_Exercise9_PCA

WEEK 4 — Day 10 Principal Component Analysis (Theory)

Contents:

1. Curse of Dimensionality
2. How to Overcome the Curse of Dimensionality?
3. Principal Component Analysis
 - 3.1. Why Do We Need PCA?
4. Understanding Principal Components
 - 4.1. What Are Principal Components?
 - 4.2 Important Properties of Principal Components
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5. How Do You Do a Principal Component Analysis?
6. Advantages
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1. Curse of Dimensionality

Curse of Dimensionality in Machine Learning arises when working with high-dimensional data, leading to increased computational complexity, overfitting, and spurious correlations.

Techniques like dimensionality reduction, feature selection, and careful model design are essential for mitigating its effects and improving algorithm performance. Navigating this challenge is crucial for unlocking the potential of high-dimensional datasets and ensuring robust machine-learning solutions.

Curse of Dimensionality refers to the phenomenon where the efficiency and effectiveness of algorithms deteriorate as the dimensionality of the data increases exponentially.

As the number of features increases, the dataset becomes high-dimensional.

High-dimensional data can lead to:

- Increased computational complexity
- Difficulty in visualization
- Redundant information
- Correlated features
- Increased storage requirements

2. How to Overcome the Curse of Dimensionality?

To overcome the curse of dimensionality, you can consider the following strategies:

Dimensionality Reduction Techniques

- **Feature Selection:** Identify and select the most relevant features from the original dataset while discarding irrelevant or redundant ones. This reduces the dimensionality of the data, simplifying the model and improving its efficiency.
- **Feature Extraction:** Transform the original high-dimensional data into a lower-dimensional space by creating new features that capture the essential information. Techniques such as Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) are commonly used for feature extraction.

3. Principal Component Analysis

Principal Component Analysis (PCA) is an unsupervised dimensionality reduction technique that transforms a large set of possibly correlated features into a smaller set of new uncorrelated variables called Principal Components.

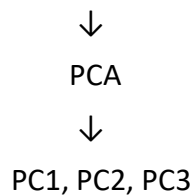
Simple Definition

PCA reduces the number of dimensions in a dataset while retaining as much of the important variation (variance) in the data as possible.

For example:

Original Dataset

X1, X2, X3, X4, X5, X13



Thus:

13 Original Features → 3 Principal Components

So, to sum up, the idea of PCA is simple: reduce the number of variables of a data set.

3.1. Why Do We Need PCA?

Suppose a dataset contains 100 features, but many of those features carry similar or correlated information.

Instead of working with all 100 features, PCA attempts to represent most of the important variation using fewer Principal Components.

100 Original Features



10 Principal Components

PCA helps in:

- Reducing the number of dimensions
- Reducing redundant information
- Reducing computational complexity
- Visualizing high-dimensional datasets
- Handling correlated features
- Simplifying subsequent analysis

4. Understanding Principal Components

4.1. What Are Principal Components?

Principal Components (PCs) are new variables that are constructed as linear combinations or mixtures of the original variables in a dataset.

For example, suppose a dataset has three original features:

X_1, X_2, X_3

PCA creates new variables:

PC_1, PC_2, PC_3

Each Principal Component is a combination of the original variables.

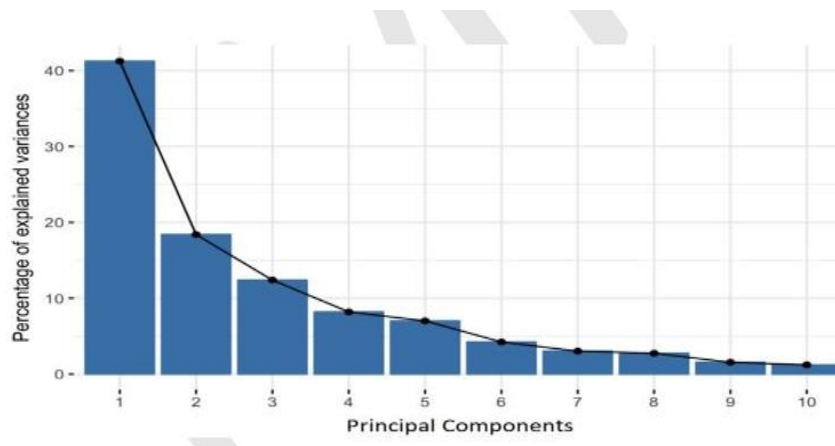
For example:

$$PC_1 = w_1X_1 + w_2X_2 + w_3X_3$$

where w_1 , w_2 , and w_3 are weights that determine how much each original feature contributes to PC_1 .

Therefore, Principal Components are new variables, not simply selected original features.

So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on, until having something like shown in the scree plot below.



Percentage of explained variance by each principal component (scree plot).

4.2 Important Properties of Principal Components

PCA creates Principal Components with two important properties:

1. Principal Components are Uncorrelated

The Principal Components are constructed so that they are uncorrelated with each other.

This means that PC_1 , PC_2 , PC_3 , etc. capture different directions of variation in the data rather than repeatedly capturing the same information.

2. Principal Components Capture Maximum Variance

PCA tries to compress or squeeze most of the important information from the original variables into the first few Principal Components.

The first component captures the maximum possible variance.

The second component captures the maximum remaining variance.

The third captures the next highest remaining variance.

And this process continues.

4.3 Geometrical Meaning of Principal Components

The geometrical interpretation makes PCA easier to understand.

Principal Components represent directions in the data that explain the maximum amount of variance.

In other words, they are directions or lines along which the data points show the greatest amount of spread.

Consider data plotted using two original features: X_1 and X_2

The original axes may not provide the best direction for representing the variation in the data.

PCA finds a new direction called PC_1 along which the data has the maximum spread.
Then it finds another direction called PC_2 , which captures the maximum remaining variance.
 PC_2 is orthogonal (perpendicular) to PC_1 .
Therefore, geometrically:
 $PC_1 \rightarrow$ Direction of maximum variance
 $PC_2 \rightarrow$ Direction of maximum remaining variance

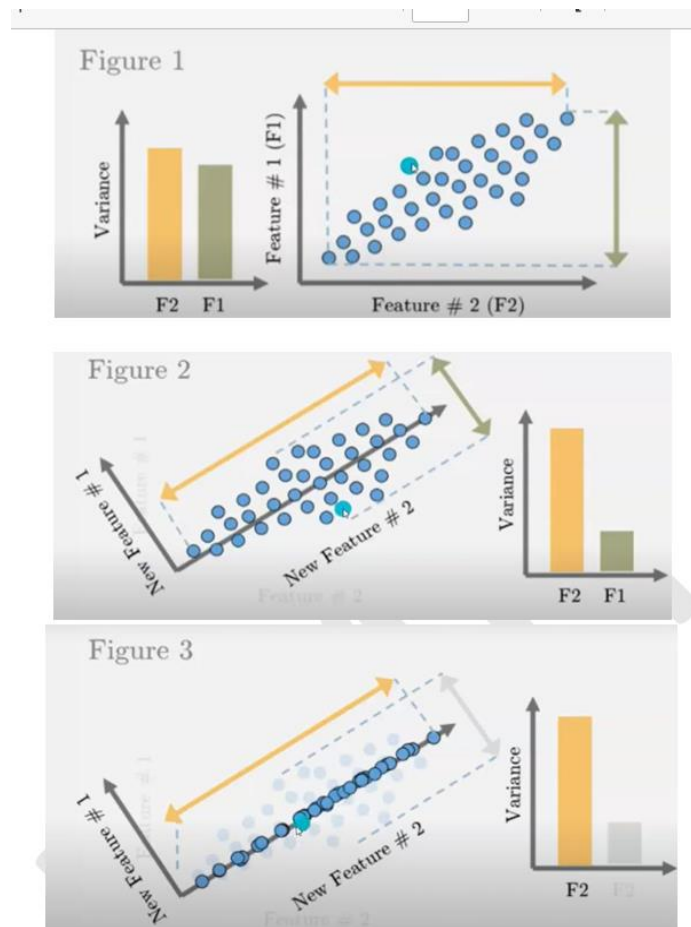


Figure 1-3: rotating the axes to the direction of maximum variance concentrates the information into New Feature #1 .

5. How Do You Do a Principal Component Analysis?

1. Standardize the range of continuous initial variables
2. Compute the covariance matrix to identify correlations
3. Compute the eigenvectors and eigenvalues of the covariance matrix to identify the principal components
4. Create a feature vector to decide which principal components to keep
5. Recast the data along the principal components axes

a. Standardization of Data

PCA is sensitive to scale; therefore, the first step is to standardize each feature so that all variables contribute equally.

That is, if there are large differences between the ranges of initial variables, those variables with larger ranges will dominate over those with small ranges (for example, a variable that ranges between 0 and 100 will dominate over a variable that ranges between 0 and 1), which will lead to biased results. So, transforming the data to comparable scales can prevent this problem.

$$z = (x - \mu) / \sigma$$

After standardization:

- Mean of each feature = 0
- Standard deviation = 1

This ensures that variables with larger numerical ranges do not dominate the analysis.

b. Compute the Covariance Matrix

The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other, or in other words, to see if there is any relationship between them.

The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables. For example, for a 3-dimensional data set with 3 variables x , y , and z , the covariance matrix is a 3×3 data matrix of this form:

$$\begin{bmatrix} \text{Cov}(x, x) & \text{Cov}(x, y) & \text{Cov}(x, z) \\ \text{Cov}(y, x) & \text{Cov}(y, y) & \text{Cov}(y, z) \\ \text{Cov}(z, x) & \text{Cov}(z, y) & \text{Cov}(z, z) \end{bmatrix}$$

Since the covariance of a variable with itself is its variance ($\text{Cov}(a, a) = \text{Var}(a)$), in the main diagonal (top left to bottom right) we actually have the variances of each initial variable. And since the covariance is commutative ($\text{Cov}(a, b) = \text{Cov}(b, a)$), the entries of the covariance matrix are symmetric with respect to the main diagonal, which means that the upper and the lower triangular portions are equal.

c. Compute Eigenvalues and Eigenvectors

Eigenvectors and eigenvalues are important linear algebra concepts used in PCA to determine the principal components of the data.

Eigenvectors and eigenvalues always occur in pairs. Each eigenvector has a corresponding eigenvalue.

The number of eigenvectors and eigenvalues is equal to the number of dimensions (features) in the dataset.

For example, if a dataset has 3 features, it is a 3-dimensional dataset. Therefore, it will have:

- 3 eigenvectors
- 3 corresponding eigenvalues

The key role of eigenvectors and eigenvalues in PCA is to identify the directions in which the data has the maximum variance (maximum information).

The eigenvectors of the covariance matrix represent the directions of the new axes. These new axes are called the Principal Components.

The eigenvalues indicate how much variance (information) is captured by each corresponding principal component.

Therefore:

Eigenvectors → Direction of the Principal Components

Eigenvalues → Amount of variance captured by the Principal Components

To identify the directions of maximum variance, we compute the eigenvalues and eigenvectors of the covariance matrix. The eigenvector associated with the largest eigenvalue represents the direction of the first principal component (PC1), which captures the maximum variance in the data.

Eigenvalue equation: $\det(\Sigma - \lambda I) = 0$

Solving this gives eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_n)$.

Next, compute eigenvectors: $(\Sigma - \lambda I)v = 0$

- Eigenvectors represent the directions of the new feature space (principal components).
- Eigenvalues indicate the amount of variance captured by each eigenvector.

The eigenvector with the highest eigenvalue becomes the first principal component (PC1).

d. Sort Eigenvalues in Descending Order

All eigenvalues are sorted from largest to smallest. The principal components are ranked based on:

- PC1: highest variance
- PC2: second highest variance
- ... and so on

This tells us which components carry the most information from the original dataset.

e. Select the Top k Principal Components

Depending on the dimensionality requirement, we select the top k eigenvectors. For example:

- 10 features → reduce to 2 components → choose eigenvectors corresponding to the top 2 eigenvalues.

The selected eigenvectors form the projection matrix,

$$W = [v_1 \ v_2 \ \dots \ v_k]$$

f. Transform the Original Dataset

The final step projects the standardized dataset onto the new feature space:

$$X_{\text{new}} = X_{\text{standardized}} \cdot W$$

This produces the dataset in reduced dimensions. After transformation:

- The features become uncorrelated
- Most variance is preserved
- The dataset becomes easier to visualize and process

6. Advantages

1. Reduces dimensionality
2. Helps visualize high-dimensional data
3. Can reduce computational complexity
4. Can reduce redundancy among correlated features
5. Creates compact data representations

7. Limitations of PCA

1. *PCA only captures linear relationships*

PCA cannot detect non-linear patterns in data since it is based on covariance and linear algebra. This makes it unsuitable for datasets with curved or complex structures.

2. *Principal components are difficult to interpret*

The new transformed features (PC1, PC2, etc.) are combinations of original variables. They do not have direct physical meaning, making interpretation challenging in real-world applications.

3. *Sensitivity to Outliers*

- Outliers can heavily influence the principal components because PCA relies on variance maximization.
- Solution: Remove or handle outliers before applying PCA, or use robust PCA variants.

4. *Loss of Information*

- PCA reduces dimensionality by discarding less significant components, which might result in the loss of important information.
- Trade-off: The balance between dimensionality reduction and retaining variance is subjective.

8. Numerical Example

Dataset (5 samples $\times$ 2 features)

Let's use this simple dataset. The goal: Reduce 2D $\rightarrow$ 1D using PCA.

Sample	X_1	X_2
1	2.5	2.4
2	0.5	0.7
3	2.2	2.9
4	1.9	2.2
5	3.1	3.0

STEP 1: Standardize / Mean-center the data

Compute means: $\bar{x}_1 = 2.04$, $\bar{x}_2 = 2.24$

Subtract the mean:

Sample	X_1'	X_2'
1	0.46	0.16
2	-1.54	-1.54
3	0.16	0.66
4	-0.14	-0.04
5	1.06	0.76

STEP 2: Covariance Matrix

The covariance between two features x_1 and x_2 is:

$$\bullet \text{ cov}(x_1, x_2) = \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{n-1}$$

Compute:

- $\text{Cov}(X_1, X_1) = 0.6165$
- $\text{Cov}(X_2, X_2) = 0.716$
- $\text{Cov}(X_1, X_2) = 0.615$

$\Sigma = (\text{covariance matrix})$

$$\begin{bmatrix} 0.6165 & 0.615 \\ 0.615 & 0.716 \end{bmatrix}$$

STEP 3: Eigenvalues & Eigenvectors

Solve the equation: $\det(\Sigma - \lambda I) = 0$

Eigenvalues:

- $\lambda_1 = 1.284$
- $\lambda_2 = 0.049$

Eigenvectors:

$\mathbf{v}_1 = [-0.6779, -0.7352]$, $\mathbf{v}_2 = [-0.7352, 0.6779]$

The higher eigenvalue = principal component direction.

STEP 4: Choose Components

To reduce from 2D $\rightarrow$ 1D, take PC1:

$\mathbf{W} = \mathbf{v}_1 = [-0.6779, -0.7352]$

STEP 5: Transform the Data

Use: $\mathbf{Z} = \mathbf{X}' \mathbf{W}$

We multiply each centered sample with PC1.

Sample	$\mathbf{X}_1'$	$\mathbf{X}_2'$	PC1 Score
1	0.46	0.16	-0.46
2	-1.54	-1.54	2.20
3	0.16	0.66	-0.77
4	-0.14	-0.04	0.15
5	1.06	0.76	-1.12

These one-dimensional values represent the PCA output.

Final PCA Result

The dataset:

- Originally 2D
- After PCA $\rightarrow$ 1D, capturing most variance in the data.

Final 1D transformed values: $[-0.46, 2.20, -0.77, 0.15, -1.12]$

9. Q & A

Q1. You are clustering images using K-Means, but the clusters look meaningless. What could improve the results?

Answer: Apply PCA before K-Means to reduce dimensionality and noise, so the distance-based clustering has fewer, more informative dimensions to work with.

Q2. A medical dataset has 200 features, and the model is slow to train and overfitting. What would you apply?

Answer: Apply PCA to reduce dimensionality while preserving maximum variance. This speeds up training and can reduce overfitting caused by irrelevant or redundant features.

Q3. Your PCA components are hard to interpret in front of stakeholders. Is this a real drawback?

Answer: Yes. PCA reduces interpretability because each component is a linear combination of the original features and does not correspond to a single, easily explainable real-world variable.

Q4. A dataset has strongly correlated features (e.g., height in cm and height in inches). How does PCA help?

Answer: PCA converts the correlated original features into a smaller number of uncorrelated principal components, removing the redundancy caused by strongly correlated inputs.

Q5. A dataset has age in years and income in lakhs. Before applying PCA, what must you do and why?

Answer: Standardize the features first. PCA maximizes variance, so a feature with a naturally larger numeric range (like income) would dominate the principal components purely due to scale, not because it is actually more informative.

Q6. After applying PCA, you find that PC1 alone explains 95% of the total variance. What does this tell you about the original features?

Answer: It suggests the original features were highly correlated and largely varied together along a single dominant direction, so most of the dataset's information can be captured in just one new dimension.

Q7. You reduce a dataset from 20 features to the top 2 principal components, and model accuracy drops slightly. Is this necessarily a problem?

Answer: Not necessarily. Some information is inevitably lost when components are discarded. A small accuracy drop is often an acceptable trade-off for faster training, simpler models, and reduced overfitting, as long as the retained components capture most of the variance.

Q8. A dataset contains several extreme outliers. You apply PCA directly on it. What issue might arise?

Answer: PCA is sensitive to outliers because it relies on maximizing variance, and outliers can heavily skew the direction of the principal components. Outliers should be handled or a robust PCA variant should be used before applying standard PCA.

Q9. One dataset has purely linear relationships between features; another has a curved, non-linear structure. Is PCA equally effective on both?

Answer: No. PCA is a linear technique based on covariance, so it works well on the linearly related dataset but fails to capture the true structure of the non-linear dataset. Techniques like Kernel PCA or t-SNE are better suited for non-linear structure.

Q10. You reduce a dataset from 50 features to 5 principal components and train a classifier. Training becomes noticeably faster and accuracy improves slightly. Why might this happen?

Answer: Reducing dimensionality removes noisy or redundant features and mitigates the curse of dimensionality, which speeds up training and can help the model generalize better by focusing on the directions that carry the most genuine signal.